

PATENTS

Unit - 2



CONTENTS

- ◉ Patents - Patentability Criteria,
- ◉ Types of patents - Process, Product & Utility Models.
- ◉ Software patenting and protection,
- ◉ Patent Infringement
 - Case studies - Apple Vs Samsung
 - Overview of patent search
 - Types of searching
- ◉ Public and Private searching databases
- ◉ Basics of patent Filing & Drafting
- ◉ Indian Patents law



Protect Your Invention

calendar

S	M	T	W	T	F	S
			1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28	29	30		

For up to

20

Years



In Specific
Territories



but NOT for

Mere Ideas

like a Time Machine

Patents can be Granted for

Products

Devices

Systems



Patents can be Granted for



Compositions

Patents can be Granted for



Processes

Methods

Uses

PATENTS

- ◉ It provides **protection** for the invention for a limited period, (**20 years** from the filing date),
- ◉ Give **permission to, or license**, other parties to use the invention on mutually agreed terms.
- ◉ He may also **sell** the right
- ◉ Once a patent expires, the protection ends, and the invention may be **used by anyone**.

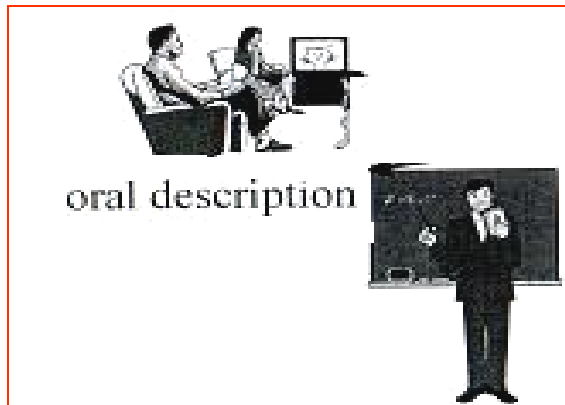
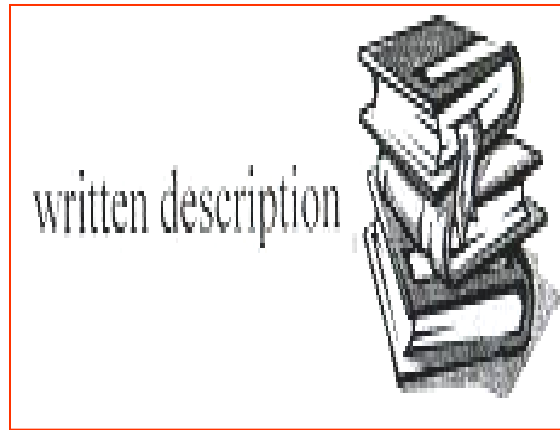
PATENTABILITY - CRITERIA

- ◉ Three criteria required for an invention to be patentable.
- ◉ Novelty
- ◉ Non Obviousness (something not early seen)-
Inventive step
- ◉ Utility

Novelty

- ⦿ **new (novel)**, meaning that the invention
 - must never have been made before,
 - carried out before or
 - used before.
- ⦿ an invention shall be considered to be new, if it does not form part of the **prior art**

PRIOR ART- EVERYTHING MADE AVAILABLE TO THE PUBLIC BY MEANS OF:



**...before the FILING DATE
of the application...**

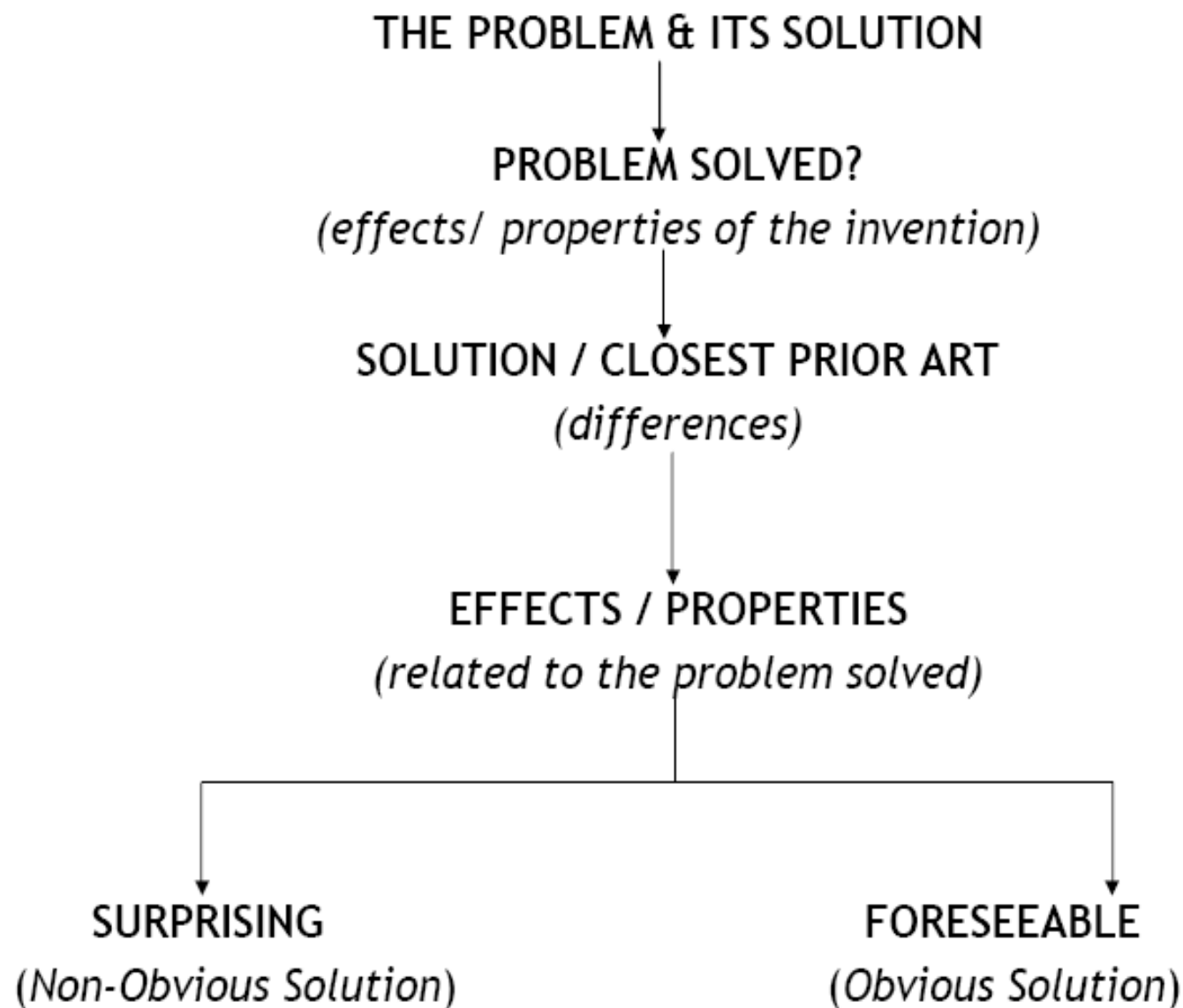
Or any other way

ON OBVIOUSNESS- INVENTIVE STEP

⊙ Inventive step is a feature of an invention that

- involves technical advance as compared to existing knowledge or
- having economic significance or
- both, making the invention non obvious to a person skilled in art.

DETERMINATION OF NON-OBVIOUSNESS/INVENTIVE STEP (PROBLEM-SOLUTION APPROACH)



- ◉ Human intervention in the inventive process.
- ◉ The term "**obvious**" means that which does not go beyond the normal progress of technology but merely follows plainly or logically from the prior art,
- ◉ i.e. something which does not involve the exercise of any skill or ability beyond that to be expected of the Person Skilled in the Art.

INDUSTRIAL UTILITY

- ◉ An invention is capable of industrial application if it satisfies three conditions, Cumulatively:
 - Can be Made
 - Can be used in at least one field of activity;
 - Can be reproduced with the same characteristics as many times as necessary
 - ie., applied for practical purposes, not be purely theoretical ;
 - intended to be a product or part of a product,
 - use it in practice.

TYPES OF PATENTS

- ◉ Process Patents
- ◉ Product Patents
- ◉ Utility Models

PROCESS PATENTS

- ◉ The patent is granted for a particular manufacturing process, and not for the product itself.
- ◉ Process patent system is preferred by the developing world.
- ◉ Any other person can produce the same product through some other PROCESS, modifying the various parameters.

- ◉ The implication is that there will be more than one producer for the same product because of the possibility of different process for the manufacturing of the product.
- ◉ Weakness of the process patent regime is that it gives less protection for the inventor.
- ◉ There is high tendency for competitors to reengineer the original invention by discovering a new process with less strain and investment.
- ◉ Benefit of process patent regime is that it reduces the element of monopoly.

PRODUCT PATENT

- ◉ The developed countries follow product patent system
- ◉ It is an exclusive right given to the original inventor of a product.
- ◉ This means that no other manufacturer can provide the same product through the same or any other process.

- ◉ The implication is that there will **not be a competitor** for the producer as it is the product which is patented.
- ◉ Product patent system gives higher level of protection to the inventor as there will not be any other patent holder.
- ◉ TRIPs follow the product patent regime.
- ◉ India's 1970 Patent Act allowed only process patent before it was amended in 2005 to comply with WTO's TRIPs provisions under which there is only product patents.

UTILITY MODEL / PETTY PATENTS

- ◉ They are sometimes referred to as "petty patents" or "innovation patents."
- ◉ Strict criteria of patentability (novelty, inventive step/non-obviousness makes it difficult for some startups
- ◉ Relaxation to upcoming industries or startups to give a boost but will also protect inventions that are useful.

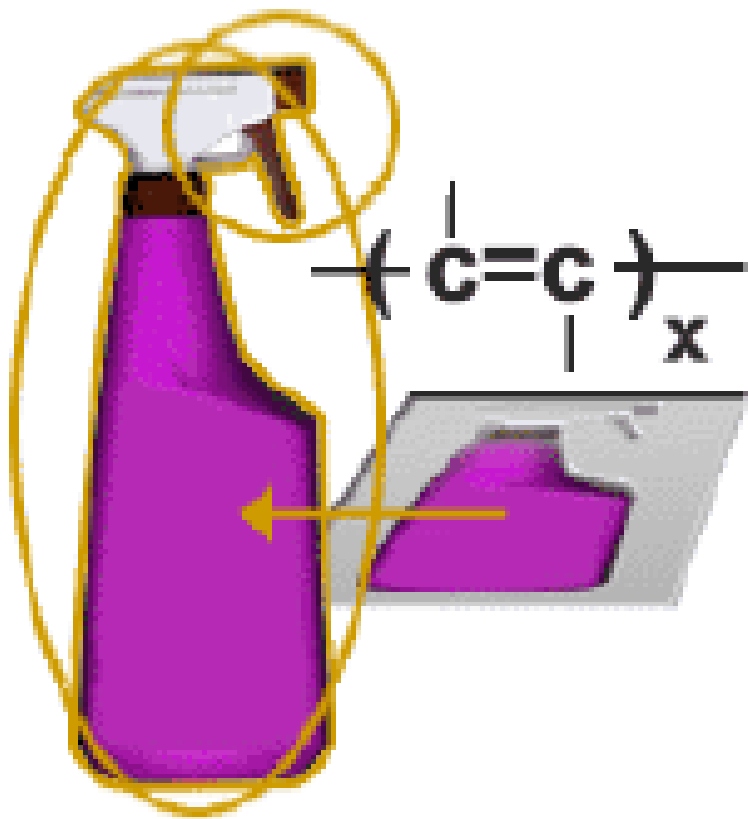
PETTY PATENTS/UTILITY MODEL

- ◉ **Novelty is a criterion** in all utility patent systems, although the standards of novelty may vary in different jurisdictions.
- ◉ **The standard of non-obviousness or inventive step** is either waived off or diluted or lower than that of patents.
- ◉ They are easier to attain than patents because
 - no strict examination is conducted before granting them,
 - making them cheaper and easily attainable.

PETTY PATENTS/UTILITY MODEL

- ◉ In most jurisdictions, there is **no substantive examination** before the grant of a utility patent, only a **preliminary procedural examination** is required.
- ◉ The duration of protection usually varies from **six to fifteen years**, as opposed to patents.

IF ONE HAS INVENTED A NEW KIND OF SPRAY BOTTLE PATENTS MAY BE GRANTED FOR



- The bottle itself (a product)
- A chemical in the plastic (chemical composition)
- The spraying mechanism (an apparatus)
- How you extruded the plastic (a process)

SOFTWARE PATENTING & PROTECTION

- Technology related patents (which are not in relation to hardware) are usually categorized as **software patents**
- software** is a collection of instructions that enable the user to interact with a computer, its hardware, to perform tasks.
- The Indian Patent Office issued revised **guidelines** on the patenting of software on February 19, 2016

- ◎ **Software** is in the form of
 - Source code ,Object code & Executable code.
- ◎ Implemented on a physical “memory” which take a lot of skill, time and labor to develop them.
- ◎ It has a market value and hence can be copied and used by unauthorized persons.

- ⊙ Patents for functional aspects of software (i.e., Processes, methods, systems, distributable code)

- ⊙ Processes that have software-driven computers as one component of larger process
- ⊙ Business method patents!

- ⊙ These should hence be protected under a strict legal regime.

Legal Protection:

- ⊙ A software invention once patented, will be valid for 20 years.

PATENT INFRINGEMENT

- ⊙ “Infringement” of a patent occurs when a competitor makes, uses, sells, offers to sell or imports an embodiment of the invention **without the permission** of the patent owner.
- ⊙ The typical remedies for infringement are:
 - ⇒ Damages (\$\$\$)
 - ⇒ Injunction (stop use by infringer)

Patent Infringement

- case studies



PATENT INFRINGEMENT : CASE STUDY

Apple
vs
Samsung



- ⦿ Accusing of infringing on software, design and technology patents related to **wireless communication technology**

APPLE vs SAMSUNG

- ◉ Samsung is a tech powerhouse: It owns key patents related to **wireless communication technology** and is the most successful Android device maker by a massive margin.
- ◉ Apple arguably *invented* the modern smartphone and tablet: It tries to protect its products with patents from “**outward design**” to “**rubber-band stretchy effect**” you get on iOS when you scroll past the edge of a page or photo.



Apple sues Samsung
for layout patent
infringement



**APRIL 15
2011**

**AUGUST
2011**

**NOVEMBER
2011**

**DEC-APRIL
2011**

**MAY
2012**

**JUNE 20
2012**

**JUNE 26
2012**

**AUGUST
2012**

**AUGUST 25
2012**

Samsung sues
Apple for
3G wireless
patent
infringement



Samsung Galaxy Tab
temporarily banned
in Australia & Europe

Unsuccessful
attempt to ban
Samsung Galaxy Tab
in the US

Australia
Germany
Netherlands
Succession of claims

Germany
Netherlands
Succession of claims

Keeps claiming that
Samsung copied
its design
Settlement talks

Still alleges wireless
patents infringement
(by Apple)

Dutch court finds
Apple guilty of
patent
infringement



Samsung Galaxy Tab 10.1
temporarily banned
in the US



Samsung releases
controversial
memo backing
Apple's claim



Apple wins \$1B
claim against
Samsung
in the US



PRODUCTS INVOLVED IN THE TRIAL:

iPhone 3G
iPhone 3GS
iPhone 4
iPad 2 3G
iPod Touch

Photos: El Tiempo/Zuma Press (Galaxy S); Apple (iPhone 4)

The Wall Street Journal
Source: WSJ research



Samsung Says...

...THE IPHONE 4 INFRINGES THESE PATENTS:

PATENT '516

Manages the phone's resources and battery by prioritizing data

PATENT '941

Provides a way to package and transmit data more efficiently

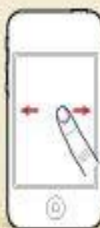


PATENT '893

Switching between pictures in a gallery and the camera

PATENT '460

Taking and emailing pictures using a scroll action



PATENT '711

A method allowing multi-tasking while playing music on the phone



Apple Says...

...THE GALAXY S INFRINGES THESE PATENTS:

PATENTS '087, '677

Ornamental design of the iPhone

PATENT '305

Rounded square icons on interface



PATENT '163

Enlarging documents by tapping the screen

SAMSUNG



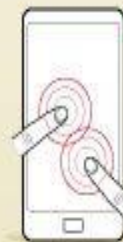
Captivate
Continuum
Droid Charge
Epic 4G
Exhibit 4G
Fascinate
Galaxy Ace
Galaxy Prevail

Galaxy S

Galaxy S 4G
Galaxy S II (AT&T)
Galaxy S II (i9100)
Galaxy S II (T-Mobile)
Galaxy S II (Epic 4G)
Galaxy S II (Skyrocket)
Galaxy S Showcase
Galaxy Tab
Galaxy Tab 10.1 (WiFi)
Galaxy Tab 10.1 (4G LTE)
Gem
Indulge
Infuse 4G
Intercept
Mesmerize
Nexus S 4G
Replenish
Transform
Vibrant

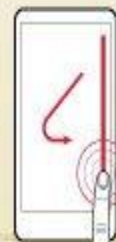
PATENT '915

Distinguishes between single-touch and multi-touch gestures



PATENT '381

'Bounce-back' feature when scrolling beyond the edge of a page



- ◉ **Apple sued Samsung**, claiming Samsung “slavishly” copied its product designs.
- ◉ Apple’s patents include design elements as well as **slide-to-unlock, rubber-banding, and universal search features**.
- ◉ Within days, **Samsung countersued** over 3G technology patents, and takes the fight international by filing claims against Apple in Japan, Germany, and Korea.
- ◉ Samsung’s complaints center on standards-essential patents for **3G mobile technology** that are **supposed to be available to anyone on fair, reasonable, and non-discriminatory (FRAND) terms**.

- ◎ The U.S. jury sides with Apple, awarding over \$1 billion in damages and finding that 26 Samsung products infringed on both Apple software and design patents including Samsung's Galaxy Tab 10.1
- ◎ Apple's \$1 billion victory over Samsung in USA
- ◎ For more info refer:
<https://www.digitaltrends.com/mobile/apple-vs-samsung-patent-war-timeline/>

ENFISH LLC vs MICROSOFT

- ◉ A DATA STORAGE AND RETRIEVAL SYSTEM FOR A COMPUTER MEMORY
- ◉ Enfish developed and sold software products, including a new type of database program.
- ◉ Microsoft develops and sells a variety of software products, including the product ADO.NET.

- ◉ Enfish's patented invention disclosed a logical model that included all data entities in a single table, with column definitions provided by rows in the same table.
- ◉ The patents describe this as the “self-referential” property of the database.
- ◉ “self-referential” :It is a specific type of data structure designed to improve the way a computer stores and retrieves data in memory.
- ◉ These include faster lookups, more efficient storage of data other than structured text.

- ◉ Enfish owned **two patents** namely, US6,151,604 and US6,163,775
- ◉ These described and claimed a **logical model for a computer database**
- ◉ A logical model is a model of data for a computer database that explains how the different elements of information are related to each other.

- ◉ In short, Enfish's patent described a table structure that allowed the information that would normally appear in several different tables to be stored in a single table.

SELF-REFERENTIAL TABLE						
ID	Type	Title	Label	Address	Employed By (#4)	Author
#1	DOCUMENT	PROJECT PLAN		C:\WORD\PROJ.DOC		#2
#2	PERSON		SCOTT WLASCHIN		#3	
#3	COMPANY		DEXIS	117 EAST COLORADO		
#4	FIELD		EMPLOYED BY			

MICROSOFT DEVELOPED A SIMILAR SOFTWARE WHICH LOOKS LIKE BELOW TABLE

OBJECT IDENTIFICATION NUMBER (OID) TO IDENTIFY ROW ,
CORRESPONDING TO A RECORD OF INFORMATION.

Document Table			
ID	Title	Address	Author
1	PROJECT PLAN	C:\WORD\PROJ.DOC	1

Document.Author --refers to--> Person.ID

Person Table		
ID	Label	Employed By
1	SCOTT WLASCHIN	1

Person.Employed_By --refers to--> Company.ID

Company Table		
ID	Label	Address
1	DEXIS	117 EAST COLORADO

- ◉ In 2012, **Enfish sued Microsoft** in the Central District of California for infringement of U.S. Patent Nos. 6,151,604 and 6,163,775.
- ◉ **Five claims in this appeal:**
 - ◉ **604 patent : claims 17, 31, & 32;**
 - ◉ **775 patent : claims 31 and 32**
- ◉ The district court, found that the Enfish claims were directed to an **ineligible abstract idea** of “storing, organizing, and retrieving memory in a logical table”.
- ◉ Without providing an inventive concept sufficient to transform the claims into a **patent-eligible invention**.

- ⦿ Enfish on **appeal** to the **Federal Circuit**, disagreed with the lower court ruling that the claims were **not patent eligible**.
- ⦿ The court found the claims to be directed to a specific improvement in the **functioning of a computer**
- ⦿ In increased flexibility, faster search times and smaller memory requirements.
- ⦿ Accordingly, the claims were found to be patent eligible

- ⦿ “Lastly, we affirm the district court’s grant of summary judgment of non-infringement and conclude that ADO.NET does not infringe claim 17”.

OVERVIEW OF PATENT SEARCH

NEED FOR PERFORMING A SEARCH



According to
Thompson Reuters

“80% of all available technical information is published in Patents and often nowhere else.”

NEED FOR PERFORMING A SEARCH



- ◉ Prior Art
- ◉ Patent - Validity / Expiration
- ◉ Infringement Analysis - to Avoid / to Sue

- ◉ Prior Art / novelty search

- ◉ A patentability search is performed to reveal prior art that can affect the patenting of an invention
- ◉ To help with the decision of applying for a patent which requires time and money
- ◉ To help with drafting the patent application in effective manner
- ◉ Both patents and publications to be considered as prior art

TYPES OF PATENT SEARCH

- ◉ Number Searching
- ◉ Name Searching
- ◉ Subject Searching
- ◉ Legal Status Searching
- ◉ IPC Classification based Searching

NUMBER SEARCHING

- ⦿ Common Format for a Patent
- ⦿ **Country Code** followed by a seven or eight digit number Eg. **GB**2013456, **WO**0213591
- ⦿ Number followed by a letter (A, B, T, etc.)
 - **Indicates the Publication Stage**
 - 'A' first stage of publication (eg US 2002-0002180 A1) &
 - 'B' Granted Patent (eg US 6,432,966 B)

◎ NAME SEARCHING

- Inventor Names
- Company and Organization Names

◎ SUBJECT SEARCHING

- **Key words** : Titles, Abstract ,
Claims
- **Classification codes are given by**
International Patent Classification

LEGAL STATUS SEARCHING

Whether,

- ⦿ Patent has been Granted or not
- ⦿ Granted Patent is still in Force
- ⦿ There are any Amendments to the Claims
- ⦿ The Ownership of patent has Changed
i.e. Assigned, Licensed, etc.

INTERNATIONAL PATENT CLASSIFICATION (IPC)

- IPC consists in a hierarchical classification system comprising:
 - Section, Classes, Subclasses, Groups (main group and subgroups)
- The IPC divides patentable technology into 8 key areas:

Section A: Human Necessities

Section B: Performing Operations, Transporting

Section C: Chemistry, Metallurgy

Section D: Textiles, Paper

Section E: Fixed Constructions

Section F: Mechanical Engineering, Lighting, Heating, Weapons

Section G: Physics

Section H: Electricity

- Follow the link to get IPC codes
<http://www.wipo.int/ipcpub/>

Example of an IPC code:



HUMAN NECESSITIES[A](#) ☐**AGRICULTURE; FORESTRY; ANIMAL HUSBANDRY; HUNTING; TRAPPING; FISHING**[A01](#) ☐**BAKING; EDIBLE DOUGHS**[A21](#) ☐**BUTCHERING; MEAT TREATMENT; PROCESSING POULTRY OR FISH**[A22](#) ☐**FOODS OR FOODSTUFFS; THEIR TREATMENT, NOT COVERED BY OTHER CLASSES**[A23](#) ☐**TOBACCO; CIGARS; CIGARETTES; SMOKERS' REQUISITES**[A24](#) ☐**WEARING APPAREL**[A41](#) ☐**HEADWEAR**[A42](#) ☐**FOOTWEAR**[A43](#) ☐**HABERDASHERY; JEWELLERY**[A44](#) ☐**HAND OR TRAVELLING ARTICLES**[A45](#) ☐**BRUSHWARE**[A46](#) ☐**FURNITURE** (arrangements of seats for, or adaptations of seats to, vehicles [B60N](#)); **DOMESTIC ARTICLES OR APPLIANCES**; COFFEE MILLS; SPICE MILLS; SUCTION CLEANERS IN GENERAL (ladders [E06C](#))[A47](#) ☐**MEDICAL OR VETERINARY SCIENCE; HYGIENE**[A61](#) ☐**LIFE-SAVING; FIRE-FIGHTING** (ladders [E06C](#))[A62](#) ☐**SPORTS; [▲ top](#) GAMES; AMUSEMENTS**[A63](#) ☐

(19) World Intellectual Property Organization
International Bureau



(43) International Publication Date
13 March 2003 (13.03.2003)

PCT

(10) International Publication Number
WO 03/020206 A3

- (51) International Patent Classification⁷: **A61K 31/7072**,
31/551
- (21) International Application Number: PCT/US02/26675
- (22) International Filing Date: 21 August 2002 (21.08.2002)
- (25) Filing Language: English
- (26) Publication Language: English
- (30) Priority Data:
60/316,745 31 August 2001 (31.08.2001) US
- (71) Applicant: **BRISTOL-MYERS SQUIBB COMPANY**
[US/US]; P.O. Box 4000, Princeton, NJ 08543-4000 (US).
- (72) Inventor: **BECHTOLD, Clifford, M.**; 17 Streamview
Circle, Guilford, CT 06437 (US).
- (74) Agent: **MORSE, David, M.**; Bristol-Myers Squibb Com-
pany, P.O. Box 4000, Princeton, NJ 08543-4000 (US).
- (81) Designated States (*national*): AE, AG, AL, AM, AT, AU,
AZ, BA, BB, BG, BR, BY, BZ, CA, CH, CN, CO, CR, CU,

CZ, DE, DK, DM, DZ, EC, EE, ES, FI, GB, GD, GE, GH,
GM, HR, HU, ID, IL, IN, IS, JP, KE, KG, KP, KR, KZ, LC,
LK, LR, LS, LT, LU, LV, MA, MD, MG, MK, MN, MW,
MX, MZ, NO, NZ, OM, PH, PL, PT, RO, RU, SD, SE, SG,
SI, SK, SL, TJ, TM, TN, TR, TT, TZ, UA, UG, UZ, VN,
YU, ZA, ZM, ZW.

- (84) Designated States (*regional*): ARIPO patent (GH, GM,
KE, LS, MW, MZ, SD, SL, SZ, TZ, UG, ZM, ZW),
Eurasian patent (AM, AZ, BY, KG, KZ, MD, RU, TJ, TM),
European patent (AT, BE, BG, CH, CY, CZ, DE, DK, EE,
ES, FI, FR, GB, GR, IE, IT, LU, MC, NL, PT, SE, SK,
TR), OAPI patent (BF, BJ, CF, CG, CI, CM, GA, GN, GQ,
GW, ML, MR, NE, SN, TD, TG).

Published:

— with international search report

- (88) Date of publication of the international search report:
2 October 2003

*For two-letter codes and other abbreviations, refer to the "Guid-
ance Notes on Codes and Abbreviations" appearing at the begin-
ning of each regular issue of the PCT Gazette.*

(54) Title: USE OF ATAZANAVIR IN HIV THERAPY

(57) Abstract: A method for reducing elevated plasma LDL and/or triglyceride levels in an HIV-infected patient is disclosed. A method, atazanavir (BMS-232632) can be used to treat HIV infection in patients exhibiting elevated plasma LDL-cholesterol and/or triglyceride levels, can be substituted for an offending HIV protease inhibitor used in such therapy, or can be used in combination with an HIV protease inhibitor metabolized by cytochrome P450 monooxygenase.

GNAS

WO 03/020206 A3

SEARCH ENGINES

Free Sites

- ◉ Google Patent Search
- ◉ USPTO
- ◉ Espacenet
- ◉ WIPO
- ◉ Free Patents Online
- ◉ Indian Patent Office
- ◉ Patentscope

Paid Sites

- STN,
- DialogPRO
- Thomson Corporation
 - Delphion
 - Patentweb
 - Aureka
 - Derwent
 - MicroPatent, etc.

For some more databases :

<http://www.wipo.int/ipdl/en/resources/links.jsp>

STEPS INVOLVED PERFORMING SEARCH

- ◉ Step 1: Understand the Invention
- ◉ Step 2: Identifying Elements of Invention/Concepts
- ◉ Step 3: Synonyms for Concepts/associated keywords
- ◉ Step 4: Identifying Class Codes
- ◉ Step 5: Search in Multiple Databases
- ◉ Step 6: Mapping

STEP 1:- UNDERSTAND THE INVENTION

- ◉ Identify Novelty, Problem/Solution from the invention disclosure document
- ◉ Understand “elements of the invention” that need to be protected and design searches accordingly

Example:

- ◉ You are looking for “warning systems allowing to continuously verifying car tire pressure”.
- ◉ You can find several concepts and associated keywords in the table below:

STEP2: IDENTIFYING ELEMENTS OF INVENTION/CONCEPTS

Concepts	Associated keywords
Tire	tire, tyre...
Pressure	pressure...
Continuous verification	verification, check, monitoring...
Warning	warning, alarm...

STEP 3: SYNONYMS FOR CONCEPTS/ASSOCIATED KEYWORDS

Patent drafter may use different terminology for the same word

Synonyms for the keywords can be searched in

- Indexed keywords-Scopus
- Thesaurus
- Pubmed Mesh-Pubmed

Note: Do not forget that some words could be written differently between UK English and US English (e.g. airplane and aeroplane), and that some letters can be substituted ("s" and "z").

STEP4: IDENTIFYING CLASS CODES

- Patent classification codes indicate the technical field or fields to which the patent application relates.
- The most used classification is the International Patent Classification (IPC).

SEARCH STRATEGY

- ◉ Class code of concept1+Synonyms of concept 2
 - ◉ Synonym of concept1+ Class code of concept 2
 - ◉ Synonym of concept1 +Synonyms of concept 2
-
- Combining Class codes and keywords for retrieving patents from multiple databases

For Ex: If two concepts are there

Concept1: tyre

Concept2: Alarm

Class code of concept 1 - A47F7/04

Synonym of concept 2 - Warning

STEP 5:- SEARCH IN MULTIPLE DATABASES

- **Google prior-art search**
 - Use Google 'prior-art search tool' to identify both patent and non-patent prior-art data
- **Keyword based patent search**
 - Google Patents, Micropat, SIP and Thomson Innovation
- **Non-patent literature search**
 - Google Scholar, IEEE search, Scirus

STEP 6: MAPPING

- ◉ Read and Analyze the patents we have retrieved relating to our innovation
- ◉ Map the patent number with elements of innovation for a better understanding

	Elements of Invention		
Patent No/Article	Tyre	Warning	Pressure
US2009349 B1			
US 8,389,492 B2			

BASICS OF PATENT DRAFTING - PARTS OF A PATENT

1. Title

2. Technical Field of the invention

3. Abstract

Within 150 words and shall commence with the title of the invention.

4. Background of the invention

Prior Art and Problem to be solved

(12) **United States Patent**
Kamo et al.

(10) **Patent No.:** **US 6,759,460 B2**
(45) **Date of Patent:** **Jul. 6, 2004**

(54) **RESIN COMPOSITION**

(75) Inventors: **Hiroshi Kamo**, Sodegaura (JP); **Shingo Kuga**, Chiba (JP); **Takashi Ono**, Yokohama (JP); **Masanori Ikeda**, Fuji (JP)

(73) Assignee: **Asahi Kasei Kabushiki Kaisha**, Osaka (JP)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

(21) Appl. No.: **10/086,852**

(22) Filed: **Mar. 4, 2002**

(65) **Prior Publication Data**

US 2002/0193533 A1 Dec. 19, 2002

(30) **Foreign Application Priority Data**

Mar. 8, 2001 (JP) 2001-064678
Sep. 6, 2001 (JP) 2001-270006

(51) **Int. Cl.**⁷ **C08K 5/34**; C08K 5/52;
C08G 65/48

(52) **U.S. Cl.** **524/100**; 524/127; 524/389;
524/399; 524/400; 524/497; 525/397; 525/437

5,997,765 A * 12/1999 Furuta et al. 252/299.01
6,322,882 B1 * 11/2001 Moritomi 428/220
6,432,323 B1 * 8/2002 Maeda et al. 252/299.01
6,518,336 B1 * 2/2003 Yabuhara et al. 524/116

FOREIGN PATENT DOCUMENTS

DE 3644208 C2 7/1987
EP 0747437 A2 11/1996
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OTHER PUBLICATIONS

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Primary Examiner—Kriellon A. Sanders

(74) *Attorney, Agent, or Firm*—Birch, Stewart, Kolasch & Birch, LLP

(57)

ABSTRACT

The present invention relates to a novel resin composition which is excellent in the balance of impact resistance, chemical resistance and fluidity as well as heat resistance and flame retardancy, and to a method for obtaining the

BACKGROUND OF THE INVENTION

1. Field of the Invention

The present invention relates to a novel resin composition which is excellent in the balance of impact resistance, chemical resistance and fluidity as well as heat resistance and flame retardancy, and to a method for obtaining the composition.

2. Prior Art

In general, polyphenylene ethers are resins that have excellent properties, for example, heat resistance, hot water resistance, size stability and mechanical and electrical properties, but are accompanied with drawbacks such as poor moldability owing to their high melt viscosity, bad chemical resistance and low impact resistance. With a view to improving such defects of polyphenylene ethers, methods of alloying and modifying polyphenylene ethers have conventionally been conducted.

It is well-known that adding polystyrene to the polyphenylene ethers improves the fluidity of the compositions. However, there are problems, which are, for example, that adding polystyrene impairs the heat resistance and chemical resistance.

As a technique related to alloying of a polyphenylene ether with another resin, for example, JP-A-56-115357 and EP-A-30417 proposed a process of mixing polymers, for example, a polyphenylene ether and a liquid-crystal polyester, thereby improving the melt processability of the polyphenylene ether. This process, however, is not sufficient to improve the impact resistance, chemical resistance and fluidity. JP-A-2-97555 proposed a process of mixing a polyphenylene ether with a liquid-crystal polyester in order to improve solder heat resistance, while JP-A-6-122762 and U.S. Pat. No. 5,498,689 proposed a process of mixing an amine-modified polyphenylene ether with a liquid-crystal polyester. Neither process is sufficient for a balance of impact resistance, chemical resistance, fluidity, heat resistance and flame retardancy. U.S. Pat. No. 5,006,403 proposed a polyphenylene ether resin composition containing fibrillated liquid-crystal polymers which form disperse phases in a continuous phase so as to reinforce the composition. WO 9902607 proposed a composition, which comprises a polyphenylene ether resin and a liquid-crystal polymer with a certain morphology. Neither process is sufficient for a balance of impact resistance, chemical resistance and fluidity.

SUMMARY OF THE INVENTION

An object of the invention is to provide a resin composition which can simultaneously attain excellent impact

1. A resin composition comprising:

(A) 70 to 99 parts by weight of a polyphenylene ether resin, and

(B) 1 to 30 parts by weight of a liquid-crystal polyester, wherein the polyphenylene ether resin in the composition contains 10 to 30 wt. % of polymer having a molecular weight of 20,000 or less and has a molecular weight distribution (Mw/Mn) of 1.8 to 3.5.

2. The resin composition according to the above item 1, wherein the polyphenylene ether resin contains 10 to 25 wt. % of polymer having a molecular weight of 20,000 or less, and has a molecular weight distribution (Mw/Mn) of 2.0 to 3.0.

3. The resin composition according to the above item 1, further comprising:

(C) 0.1 to 10 parts by weight of a compound containing a monovalent, divalent, trivalent or tetravalent metal element based on 100 parts by weight, in total, of components (A) and (B).

4. The resin composition according to the above item 3, wherein the monovalent, divalent, trivalent or tetravalent metal element is at least one element selected from the group consisting of Zn, Mg, Ti, Sn, Sb, Al and Ge.

5. The resin composition according to the above item 3, wherein component (C) is at least one compound selected from the group consisting of ZnO, zinc acetate, zinc stearate, Mg(OH)₂, tetrabutoxide titanate and tetraisopropoxy titanate.

6. The resin composition according to any one of the above items 3 to 5, wherein the composition has a morphology comprising a continuous phase and a disperse phase, in which the abundance ratio (R) of Zn and/or Mg in the disperse phase obtained by TEM-EDX is 0.0005 or more and a requirement of $R_d > R_m$ when $R_m = 0$ or $150 R_m > R_d > R_m$ when $R_m \neq 0$ (wherein the abundance ratio (R) of Zn and/or Mg = (the number of L α -rays of Zn and/or Mg)/(the number of K α -rays of C), R_d represents R in the disperse phase, and R_m represents R in the continuous phase) is satisfied.

7. The resin composition according to any one of the above items 1 to 5, further comprising:

(D) 0.05 to 30 parts by weight of a vinyl compound elastomer based on 100 parts by weight, in total, of components (A) and (B).

8. The resin composition according to the above item 7, wherein component (D) is functionalized with an acid anhydride group.

9. The resin composition according to any one of the above items 1 to 5, further comprising:

(E) 0.1 to 10 parts by weight of a flame retardant based on 100 parts by weight, in total, of components (A) and

PARTS OF A PATENT DRAFTING APPLICATION

5. Summary of the invention

Various aspects of the invention and the principles on which it is based

6. Brief description of the Drawings

7. Detailed description of the invention

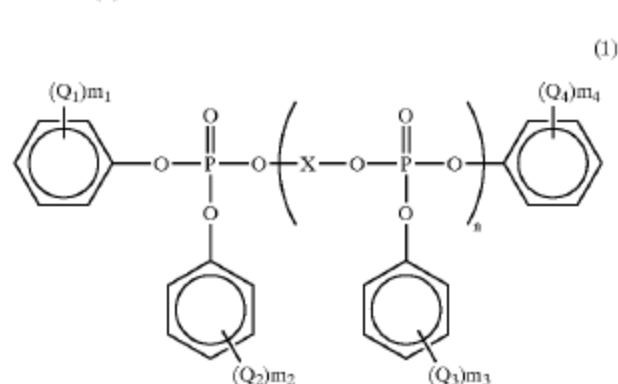
A few Industrial Applications- at the end

8. Claims

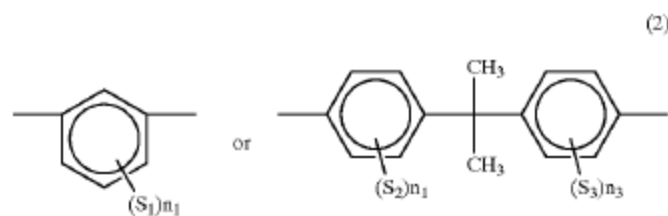
Extent of protection

3

- (G) a cyclic nitrogen compound, wherein the weight ratio (f/g) of component (F) to component (G) is 0.1 to 10.
15. The resin composition according to the above item 14, wherein component (G) is a melamine, melem or mello.
16. The resin composition according to the above item 9, wherein component (E) is a phosphorus flame retardant.
17. The resin composition according to the above item 16, wherein the phosphorus flame retardant has the following structure (1):



wherein Q_1 , Q_2 , Q_3 and Q_4 each independently represents an alkyl group having 1 to 6 carbon atoms or a hydrogen atom; n represents an integer of 1 or more; m_1 , m_2 , m_3 and m_4 each independently represents an integer of 0 to 3; and X is selected from formula (2):



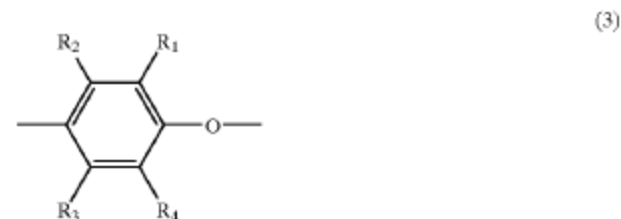
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wherein the abundance ratio (R) of Zn and/or Mg in the disperse phase obtained by TEM-EDX is 0.0005 or more, and a requirement of $R_d > R_m$ when $R_m = 0$ or $150 R_m > R_d > R_m$ when $R_m \neq 0$ (wherein the abundance ratio (R) of Zn and/or Mg = (the number of L α -rays of Zn and/or Mg)/(the number of K α -rays of C), R_d represents R in the disperse phase, and R_m represents R in the continuous phase) is satisfied.

DETAILED DESCRIPTION OF THE INVENTION

The present invention is described more specifically below.

The polyphenylene ether (A) is a homopolymer and/or copolymer comprising a recurring unit represented by the following formula (3):



(wherein R_1 and R_4 each independently represents a hydrogen atom, a primary or secondary lower alkyl group, a phenyl group, an aminoalkyl group or a hydrocarbon oxy group; and R_2 and R_3 each independently represents a hydrogen atom, a primary or secondary lower alkyl group or a phenyl group), and having a reduced viscosity (as measured under the condition: 0.5 g/dl, chloroform solution, and 30° C.) of 0.15 to 1.0 dl/g. The reduced viscosity is preferably within a range of 0.20 to 0.70 dl/g, more preferably within a range of 0.40 to 0.60 dl/g.

Specific examples of the polyphenylene ether include

What is claimed is:

1. A resin composition comprising:

(A) 70 to 99 parts by weight of a polyphenylene ether resin, and

(B) 1 to 30 parts by weight of a liquid-crystal polyester, wherein the polyphenylene ether resin in the composition contains 10 to 30 wt. % of polymer having a molecular weight of 20,000 or less and has a molecular weight distribution (Mw/Mn) of 1.8 to 3.5.

2. The resin composition according to claim 1, wherein the polyphenylene ether resin contains 10 to 25 wt. % of polymer having a molecular weight of 20,000 or less, and has a molecular weight distribution (Mw/Mn) of 2.0 to 3.0.

3. The resin composition according to claim 1, further comprising:

(C) 0.1 to 10 parts by weight of a compound containing a monovalent, divalent, trivalent or tetravalent metal element based on 100 parts by weight, in total, of components (A) and (B).

4. The resin composition according to claim 3, wherein the monovalent, divalent, trivalent or tetravalent metal element is at least one element selected from the group consisting of Zn, Mg, Ti, Sn, Sb, Al and Ge.

5. The resin composition according to claim 3, wherein component (C) is at least one compound selected from the group consisting of ZnO, zinc acetate, zinc stearate, Mg(OH)₂, tetrabutoxide titanate and tetraisopropoxy titanate.

6. A resin composition comprising:

(A) 70 to 99 parts by weight of a polyphenylene ether resin;

(B) 1 to 30 parts by weight of a liquid-crystal polyester, wherein the polyphenylene ether resin in the composition

abundance ratio (R) of Zn and/or Mg in the disperse phase obtained by TEM-EDX is 0.0005 or more and a requirement of $R_d > R_m$ when $R_m = 0$ or $150 R_m > R_d > R_m$ when $R_m \neq 0$ (wherein the abundance ratio (R) of Zn and/or Mg = (the number of L α -rays of Zn and/or Mg)/(the number of K α -rays of C), R_d represents R in the disperse phase, and R_m represents R in the continuous phase) is satisfied.

7. The resin composition according to any one of claims 1 to 5, further comprising:

(D) 0.05 to 30 parts by weight of a vinyl compound elastomer based on 100 parts by weight, in total, of components (A) and (B).

8. The resin composition according to claim 7, wherein component (D) is functionalized with an acid anhydride group.

9. The resin composition according to any one of claims 1 to 5, further comprising:

(E) 0.1 to 10 parts by weight of a flame retardant based on 100 parts by weight, in total, of components (A) and (B).

10. The resin composition according to claim 9, wherein component (E) is (F) a silicon compound.

11. The resin composition according to claim 10, wherein component (F) is a silicone.

12. The resin composition according to claim 10, wherein component (F) is a polyhedral oligomeric silsesquioxane or partially opened polyhedral oligomeric silsesquioxane.

13. The resin composition according to claim 10, wherein component (F) is a silica.

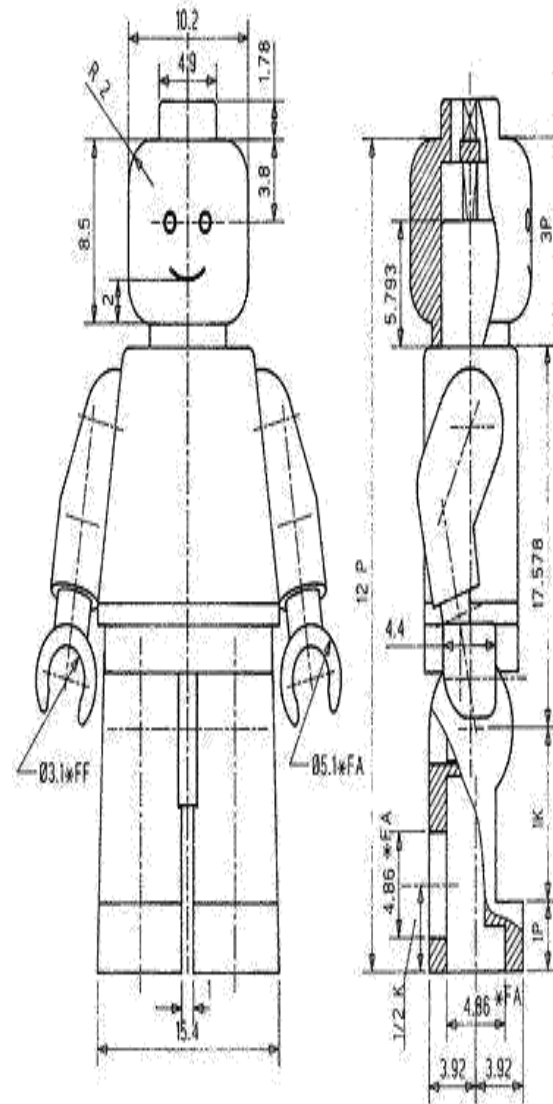
PATENT DRAFTING

- Patent is techno-legal document containing scientific information
- It includes specifications and Claims

Specification

Detailed description of invention and background

Drawings - diagrams, flow charts, data



Claims -Heart of the patent

- ◉ Define the invention for which patent protection is being sought, legally
- ◉ Provide a sufficient disclosure of the invention that a skilled person in the relevant field could make & understand the invention,
- ◉ Include both broad and narrow claims
- ◉ Claims are classified as **dependent and independent claims**

- ⦿ An independent claim is the broadest claim.
- ⦿ Each dependent claim refers to an independent claim and includes all of its features, and then adds further detail to the independent claim.
- ⦿ Each dependent claim is narrower than the independent claim

LET'S TAKE A LOOK AT SOME SAMPLE CLAIMS TO SEE

1. **A method comprising steps A, B, and C.**
2. The **method of Claim 1**,
further comprising step D.
3. The **method of Claim 2**,
further comprising step E.
4. The **method of Claim 1**,
further comprising step F.
5. **A method comprising steps A, B, and G.**
6. The **method of Claim 5**,
further comprising step C.

PATENT FILING

- ◉ File an application for patent
 - With one of the patent offices based on territorial jurisdiction of the place of office or residence of the applicant /agent
 - Pay the required fee

- ◉ Information concerning application form and details of fee available at www.ipindia.nic.in

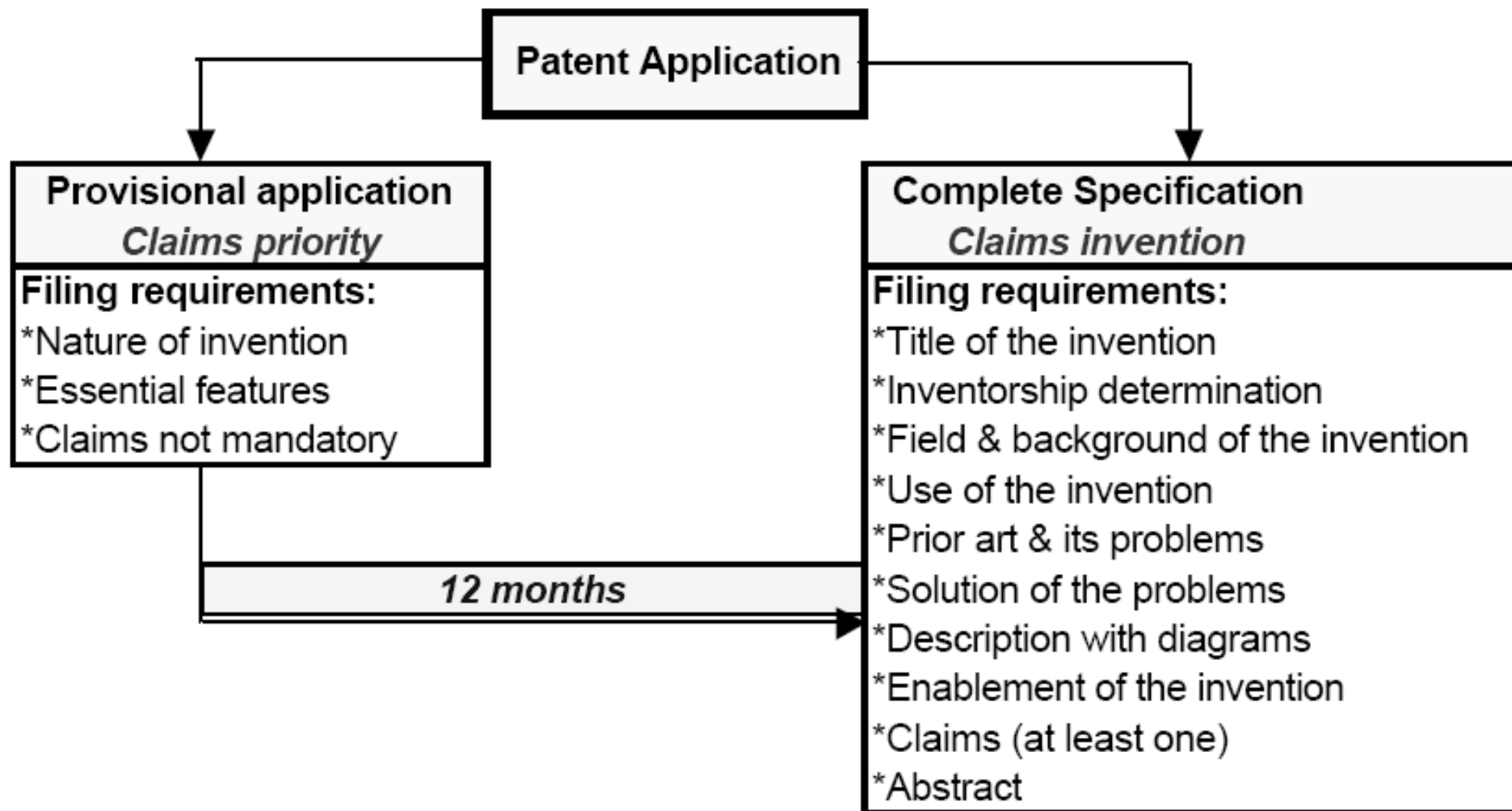
TYPES OF PATENT APPLICATION

- The first step in securing a patent is the filing of a patent application.
- Applications may be provisional or complete
- ✓ **Provisional specification**
- ✓ **Complete specification**

- If the invention is still in the development mode and tests are underway, it is a good idea to quickly file a provisional application to block the all-important filing date.

- Filing of the provisional application gives you 12 months of time to test and finalize your invention and file the complete application

Filing of Patent Application in India



- ◉ *Foreign filing decision* - Further, if you are interested in protecting your invention in foreign jurisdictions, the maximum time allowed is 12 months from your first filing date.
- ◉ Based on the countries you are interested in; you can opt for filing a **convention application in Paris convention members** individually in each of the countries you are interested in protecting your invention.
- ◉ Alternatively, you can use the **Patent Cooperation Treaty (PCT) system** to **reserve your right in 140 odd member countries.**

FORMALITY CHECK

- ⦿ An Examiner checks the formal requirements before accepting the application and the fee - this is done immediately
- ⦿ Issue of application number and the cash receipt - this is done the same day

PUBLICATION

- ◉ Every patent application which is filed with the Indian patent office is kept as a secret until the time it is published in the official patent journal.
- ◉ Indian patent office will publish patent applications ordinarily after 18 months. This is an automatic event and you need not make any request.
- ◉ However, if you wish to get your application published earlier, you can make a request for early publication (Form 9) and your application will ordinarily be published within 1 month of the request.

EXAMINATION OF THE PATENT APPLICATION

- ◉ Every patent application which is filed for protection has to be substantively examined before a patent is finally granted.
- ◉ The examination process is where your patent application will finally be examined on merits of the invention as described and claimed in the patent specification.

- ◉ Application is examined on request-Form 18
- ◉ A period of 48 months, from the date of filing, is available for making request for examination
- ◉ Hence, if you wish to expedite the patenting process, filing of form 9 and 18A at an early stage is advised.
- ◉ The fee for this is INR 8000. At present, the patent office has limited this request to about 1000 request in a year.

- ⊙ Application is sent to an Examiner within 1 month from the date of request for examination
- ⊙ Examiner undertakes examination w.r.t.
 - whether the claimed invention is not prohibited for grant of patent
 - whether the invention meets the criteria of patentability

ISSUE OF FER

- ◉ During the examination process, the examiner will scrutinize the application to ensure that the application is in accordance with the patent act and rules.
- ◉ The examiner also performs a search to understand similar technologies to ascertain if the invention would satisfy the patentability criteria.
- ◉ Based on the review of the application, the examiner will issue an Examination Report to the applicant, stating the grounds for objections.
- ◉ The first such examination report is called the First Examination Report (FER).

RESPONSE FROM THE APPLICANT

- ⦿ 12 months' time, from the date of issue of FER, is available to the applicant to meet the objections
- ⦿ If objections are met, grant of patent is approved by the Controller - within a period of 1 month

PRE-GRANT OPPOSITION

- ⦿ After publication, an opposition can be filed within a period of 6 months
- ⦿ Opportunity of hearing the opponent is also available

CONSIDERATION OF PRE-GRANT OPPOSITION

- After examining the opposition and the submissions made during the hearing, Controller may
 - Either reject the opposition and grant the patent
 - Or accept the opposition and modify/reject the patent application
- This is to be done within a period of 1 month from the date of completion of opposition proceedings

GRANT OF A PATENT

- ⦿ A certificate of patent is issued within 7 days
- ⦿ Grant of patent is published in the official journal

RENEWAL

- ◉ After the patent has been granted, it has to be renewed every year by paying the renewal fee.
- ◉ A patent in India can be renewed for a maximum period of 20 years from the patent filing date.

STAGES - FILING TO GRANT OF PATENT

**FILING OF APPLICATION
PROVNL. / COMPLETE**

• IF P.S. IS FILED C.S. TO BE FILED WITHIN 12 MONTHS

PUBLICATION OF APPLICATION

• PROMPTLY AFTER 18 MONTHS FROM P.D.

REQUEST FOR EXAMINATION

• WITHIN 48 MONTHS FROM F.D.

EXAMINATION-ISSUE OF FER

GRANT OF PATENT

OPPOSITION

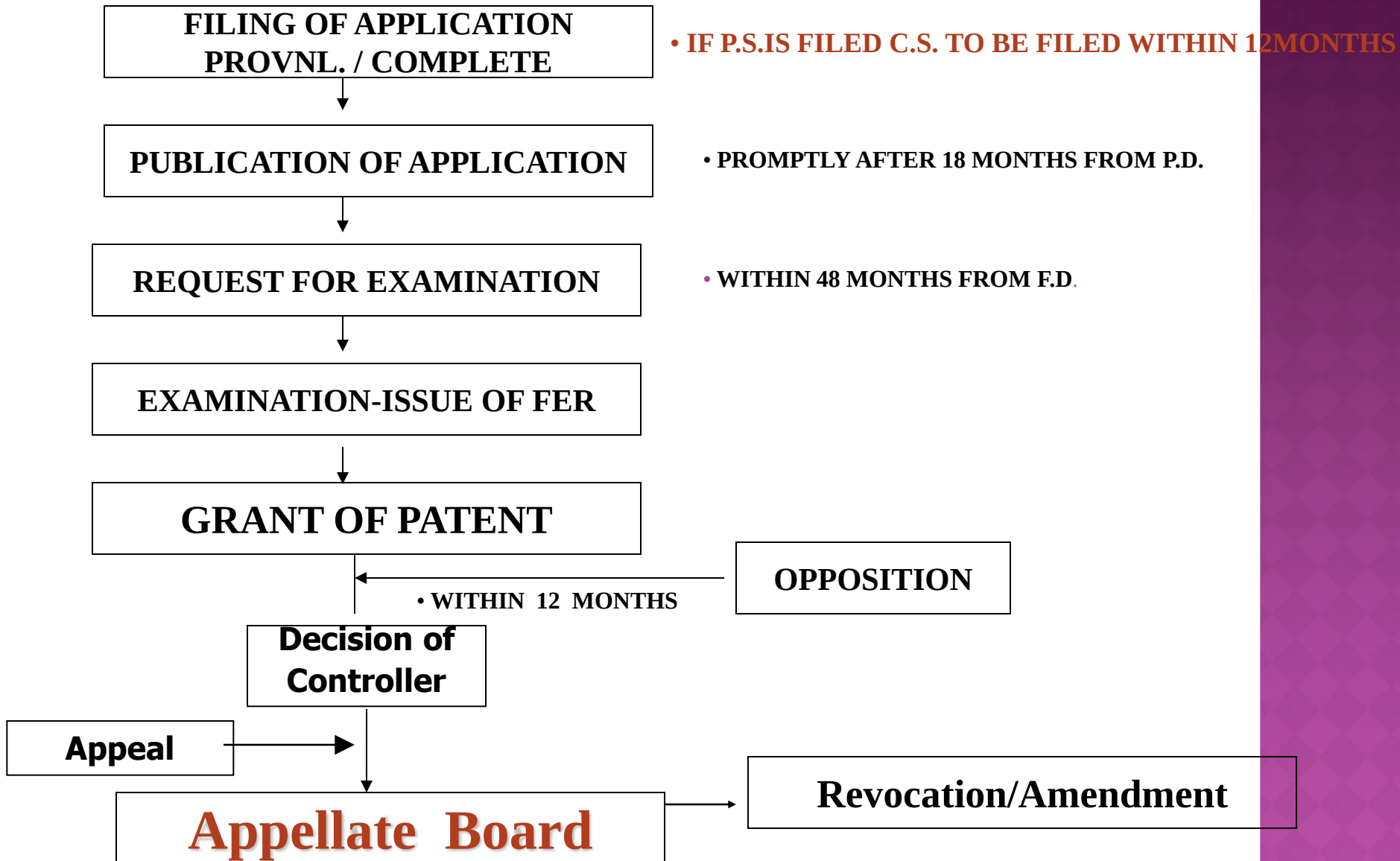
• WITHIN 12 MONTHS

**Decision of
Controller**

Appeal

Appellate Board

Revocation/Amendment



PATENT LAW AND REGULATIONS

○ Patents Act, 1970

■ Amended in

- 1999

- 2002

- 2005

AMENDMENTS

◉ **From 1999**

- Mail-Box for pharmaceutical and agrochemicals products
- Exclusive Marketing Rights

◉ **From 2002**

◉ Patent term increased to 20 years

- Definition of invention - inclusion of inventive step
- Reversal of burden of proof - on the infringer
- Mandatory compulsory license provision for food, drugs and chemicals removed
- Right of patentee (importation also included)

◉ **From 2005**

- Product patents for food, chemical and pharmaceutical

We have met our international commitments

INDIAN PATENT ACT

- ◉ THE PATENTS ACT 1970 (Updated till 2005 amendment)
- ◉ Chapter I - Section 1- 2 - Definitions and interpretation
- ◉ Chapter II - Inventions Not Patentable Section 3 - 5
- ◉ Chapter III - Applications for Patents, Section 6 -11
- ◉ Chapter IV - Publication and Examination of Applications Section 12 - 24
- ◉ Chapter V - Opposition Proceedings to Grant of Patents Section 25 - 28
- ◉ Chapter VI - Anticipation Section 29 - 34
- ◉ Chapter VII - Provisions for Secrecy of Certain Inventions Section 35 - 42
- ◉ Chapter VIII - Grant of Patents and Rights conferred thereby Section 43 - 53
- ◉ Chapter IX - Patents of Addition Section 54 - 56
- ◉ Chapter X - Amendment of Applications and Specification Section 57 - 59
- ◉ Chapter XI - Restoration of Lapsed Patents Section 60 - 62
- ◉ Chapter XII - Surrender and Revocation of Patents
- ◉ till 23 chapters with 163 Sections.

PATENT LAW - SALIENT FEATURES

- ◉ Both product and process patent provided
- ◉ Term of patent - 20 years
- ◉ Examination on request
- ◉ Both pre-grant and post-grant opposition
- ◉ Fast track mechanism for disposal of appeals
- ◉ Provision for protection of bio-diversity and traditional knowledge
- ◉ Publication of applications after 18 months with facility for early publication
- ◉ Substantially reduced time-lines